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A fluorescence turn-off chemosensor based on N-doped carbon quantum dots for detection of Fe^{3+} in aqueous solution

Yuliang Jiang [a], Qiaorong Han [a], Can Jin [b], Jing Zhang [a], Bingxiang Wang [a,*]

[a] School of Chemistry and Materials Science, Nanjing Normal University, Nanjing 210097, China
[b] Jiangsu Provincial Key Laboratory of Biomass Energy and Materials, National Engineering Laboratory for Biomass Chemical Utilization, Institute of Chemical Industry of Forest Products, CAF, Nanjing 210042, China

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ABSTRACT

In this study, new carbon nanomaterial N-doped carbon dots (NCDs) were obtained via a one-step hydrothermal method using isoleucine and citric acid as carbon precursors. TEM results showed that the NCD diameters were not uniform but distributed in the 6–15 nm range, with an average value of 10 nm. The maximum emission wavelength of the obtained NCDs was 415 nm with an excitation wavelength of 370 nm. Under optimal conditions, the nanoparticle can be used as sensors and exhibited a 'turn-off' fluorescence response to Fe^{3+}, with a good linear relationship of (F/F_0) versus Fe^{3+} over the concentration range of 0–20 μmol/L.

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1. Introduction

The design and development of fluorescence sensors for the detection of Fe^{3+} is becoming an important issue because of the vital roles Fe^{3+} play in environmental, biological, and chemical systems [1–5]. Recently, a large number of fluorescent chemosensors were exploited as useful tools for the detection of Fe^{3+} owing to their intrinsic sensitivity, selectivity, and capacity for rapid, real-time monitoring [6–10]. Most of these chemosensors were organic dyes with environmental unfriendly, time-consuming, and cytotoxic chemical syntheses [11–14], which restricted their practical performance. Therefore, the development of an efficient low-toxicity, high-efficiency, and stable chemical probe is urgent.

Carbon dots (CDs), as the newest class of carbon nanomaterials, have attracted a great deal of attention for their fascinating properties which include size- and excitation-dependent photoluminescence (PL), low toxicity, good stability, and easy preparation; they are also nonblinking and photostable [15–20]. Moreover, some shortcomings such as low quantum yields and emission of only blue light hinder their applications. Thus, it is of great significance to take advantage of NCDs through surface modification of CDs to improve their luminescent properties for the effective detection of various metal ions.

Herein, we report a green and facile preparative route to luminescent NCDs by a direct hydrothermal method using isoleucine and citric acid (CA). The NCDs can be used for 'turn-off' detection of Fe^{3+} with high selectivity and sensitivity. The preparation procedures of NCDs are shown in Scheme 1.

2. Experimental

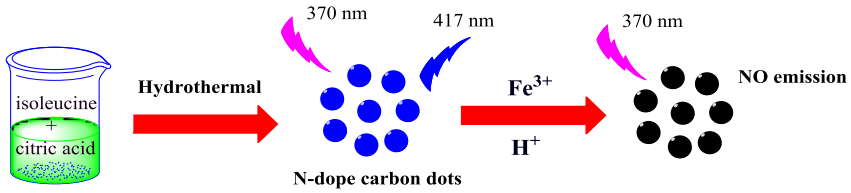
Materials: Citric acid monohydrate and isoleucine were purchased from Aladdin chemistry Co. Ltd. (Shanghai, China), and were utilized without further purification. All the other chemicals used in this paper were obtained from commercial suppliers and used directly untreated. Inorganic salts $MnCl_2$, $CaCl_2$, $ZnCl_2$, $AlCl_3$, $NaNO_3$, KNO_3, $Fe(NO_3)_3 \cdot 6H_2O$, $Co(NO_3)_2$, $Mg(NO_3)_2 \cdot 6H_2O$, $Cu(NO_3)_2 \cdot 3H_2O$, $Cd(NO_3)_2 \cdot 2H_2O$, $NiNO_3 \cdot 6H_2O$, $AgNO_3$ and Hg $(NO_3)_2 \cdot 0.5H_2O$ were dissolved in double distilled water (100 mL) to afford 1.0×10^{-3} mol/L aqueous solutions. All samples were prepared at room temperature.

Characterization: UV–vis absorption spectra were measured with a Varian Cary 50spectrophotometer. The IR spectra of the product were measured by a Nicolet Nexus 670 Fourier transform infrared (FT-IR) spectroscope. The microstructures of NCDs were examined using a transmission electron microscopy (TEM), which were recorded with a JEOL JEM-2100. Fluorescence spectra were recorded on a Varian cary Eclipse fluorescence spectrophotometer. Xe lamp was used as an excitation source. X-ray photoelectron spectroscopy (XPS) analysis was carried out on a PHI 5000 Versa Probe electron spectrometer from ULVAC-PHI.

Preparation of NCDs: Typical preparation of NCDs was described as follows: briefly, CA monohydrate (0.21 g, 1.0 mmol) and isoleucine

* Corresponding author. Tel.: +86 025 85891705; fax: +86 025 85891767.
E-mail address: wangbingxiang@njnu.edu.cn (B. Wang).

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Scheme 1. Cartoon illustrations for the synthesis process of NCDs.

(0.28 g, 1.0 mmol) were dissolved in water (5 mL) and heated hydrothermally in a Teflon-equipped stainless-steel autoclave at 180 °C for 5 h. After cooling to room temperature the brown syrup product was neutralized with 1 mol/L NaOH solution. The large particles were removed by filtration using a 0.22 μm filter membrane and then centrifugated at 12,000 rpm for 15 min. The obtained NCDs were further dialyzed in a dialysis bag (retained molecular weight: 1000 Da) for three days. The blue fluorescent NCDs were obtained.

Principles of fluorescence quenching: Dynamic quenching and static quenching are two basic forms of fluorescence quenching. The following formulas 1 and 2 represent the corresponding Stern–Volmer equation. For Eq. (1) F_0 and F are the fluorescence intensities before and after the addition of the quencher respectively, K_q is the rate constant of dynamic quenching, τ_0 is the fluorescence lifetime in the absence of the quenchers; τ is the fluorescence lifetime in the presence of quenchers, and [Q] is the quencher concentration in solution.

$$\frac{F_0}{F}=\frac{\tau_0}{\tau}=1+k_q\tau_0[Q] \quad (1)$$

$$\frac{F_0}{F}=1+k_{SV}[Q] \quad (2)$$

Static quenching occurs as a result of the formation of a non-fluorescent complex between the ground state of fluorophore and quencher in Eq. (2) the K_{SV} is now the association constant K_S. While the life-time of the fluorophore is not affected by the static quenching, lifetime measurements are a definitive method to distinguish between static and dynamic quenching as reported [21].

3. Results and discussion

The UV–vis absorption spectra of NCDs dispersed in water (Fig. S1) shown two typical absorption peaks at 242 and 343 nm, the two peaks are ascribed to π–π* transition of the C=C bond and n–π* transition of the C=O bond, respectively, which suggests that the honeycomb frame was obtained through hydrothermal procedure.

The structure and composition of NCDs were characterized by FT-IR spectroscopy. As shown in Fig. S2, the NCDs show a main absorption band of O–H at 3430 cm^{-1}, C–H at 2970 and 2850 cm^{-1}, stretching vibration of C=O at 1570 cm^{-1}, the typical band of the amine C–N(III) bond is at 1430 cm^{-1}, these results showed that amino acid as nitrogen source doped carbon quantum dots successfully and the NCDs are rich in carboxylic groups, which make it as high water solubility.

From TEM we can find that the NCDs are highly dispersed in aqueous solution and their diameters are not of uniform size distributed in the range of 6–15 nm with an average diameter of 10 nm (Fig. 1).

The composition of NCDs was carried out by X-ray photoelectron spectroscopy (XPS) spectra. The full scan XPS spectrum of NCDs (Fig. S3a) clearly reveals that carbon, nitrogen and oxygen are present at the surface of NCDs, as shown in Fig. S3b 1.14% of nitrogen and 62.79% of carbon indicate that the NCDs are nitrogen-doped and carbon-rich.

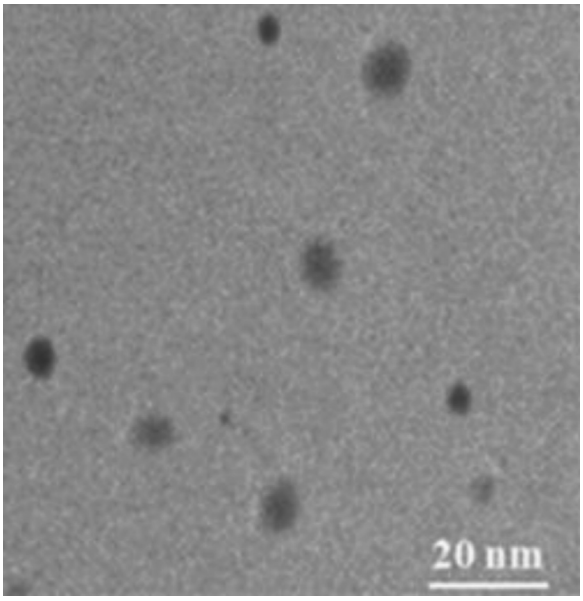


Fig. 1. TEM image of NCDs.

The response of NCDs in the presence of Fe^{3+} was determined by titration of the NCDs with Fe^{3+} in phosphate buffer solution (pH=6.8). Upon addition of Fe^{3+} at a 0–60 μM concentration, the fluorescence intensity of the NCD solution was quenched by 93%. The color of the solution also changed from blue to colorless, as detected visually under a 365 nm UV lamp (Fig. 2a). From Job's plot analysis, we found a good linear relationship of (F/F_0) versus the concentration of Fe^{3+} over a range of 0–20 μM, with a correlation coefficient of 0.98 (Fig. 2b), which confirmed previous results that Fe^{3+} interacts with NCDs by chelating with the oxygen functional groups on the surface of NCDs to result in fluorescence quenching. Under the same conditions, the addition of other anions, including Na^+, Ni^{2+}, Co^{2+}, Ag^+, Mn^{2+}, Cd^{2+}, Pb^{2+}, Hg^{2+}, K^+, Zn^{2+}, Al^{3+}, Cu^{2+}, and Fe^{2+}, resulted in small or no obvious fluorescence changes (Table 1). These findings validate the high sensitivity and selectivity of the NCDs toward Fe^{3+}.

To further study the mechanism of the quenching process, the fluorescence decays were used to test the process of NCDs in the presence and absence of Fe^{3+} ions. Fig. 3 shows the fluorescence decays of NCDs in the absence and presence of Fe^{3+}. Lifetime study shows that before and after introducing Fe^{3+} to the aqueous solution, the lifetime of NCDs exhibit obvious shift from 5.7 ns to 4.5 ns which suggests that the mechanism of the quenching process is dynamic quenching.

4. Conclusion

In summary, we have successfully synthesized for the first time a novel 'turn-off' fluorescent probe based on NCDs by the facile, green, one-step hydrothermal treatment of isoleucine and citric acid. The NCDs showed high sensitivity and selectivity for Fe^{3+}

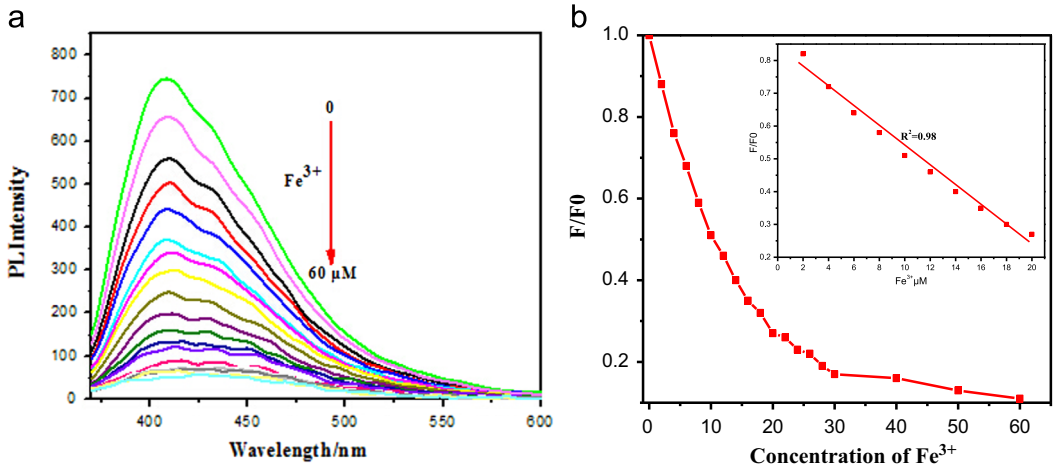


Fig. 2. (a) The fluorescence spectra of NCDs (1.0×10^{-3} M) upon the addition of different concentrations of Fe^{3+} (0, 1, 2, 4, 6, 8, 10, 15, 20, 25, 30, 35, 40, 50, and 60 μM). (b) A linear relationship of (F/F_0) versus the concentration of Fe^{3+} over the range from 10 to 60 μM. (For interpretation of the reference to color in this figure, the reader is referred to the web version of this article.)

Table 1
The fluorescence intensities of NCDs with 50 μM metal ion (λ_{ex}=370 nm, silt=5 × 5).

pH	Ni^{2+}	Co^{2+}	Ag^{+}	Mn^{2+}	Na^{+}	Cd^{2+}	Pb^{2+}	Hg^{2+}	K^{+}	Zn^{2+}	Al^{3+}	Cu^{2+}	Fe^{2+}
λ_{em} (nm)	415	415	417	418	415	415	417	418	415	416	415	415	416
I_F (a.u.)	430	437	428	403	398	418	399	423	389	388	410	415	378

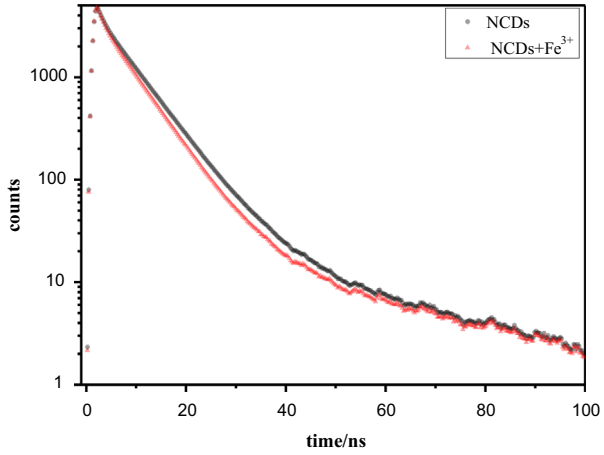


Fig. 3. Fluorescence decay curves of NCDs in the absence and presence of Fe^{3+}.

relative to other competitive metal ions such as Na^{+}, Ni^{2+}, Co^{2+}, Ag^{+}, Mn^{2+}, Cd^{2+}, Pb^{2+}, Hg^{2+}, K^{+}, Zn^{2+}, Al^{3+}, Cu^{2+}, and Fe^{2+}, by reducing the fluorescence emission at 415 nm. The fluorescence decay of the NCDs in the presence of Fe^{3+} revealed that the mechanism of quenching involves complexation and that dynamic quenching dominates over static quenching.

Acknowledgments

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Appendix A. Supplementary materials

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.matlet.2014.10.168.

References

[1] Mancini PME, Ochando LE, Parra M. Tetrahedron 2012;68:4882–7.
[2] Rathinam B, Chien CC, Chen BC, Liu JH. Tetrahedron 2013;69:235–41.
[3] Weerasinghe AJ, Abebe FA, Sinn E. Tetrahedron Lett 2011;52:5648–51.
[4] Hu ZQ, Wang XM, Feng YC, Ding L, Li M, Lin CS. Chem Commun 2011;47:1622–4.
[5] Dornelles AS, Garcia VA, De Lima MN, Vedana G, Alcalde LA, Bogo MR, Schroeder N. Neurochem Res 2010;35 (564–564).
[6] Mei QS, Jiang CL, Guan GJ, Zhang K, Liu BH, Liu RY, Zhang ZP. Chem Commun 2012;48:7468–70.
[7] Shellaiah M, Wu YH, Singh A, Lin HC. J Mater Chem A 2013;1:1310–8.
[8] Ding YJ, Zhu H, Zhang XX, Zhu JJ, Burda C. Chem Commun 2013;49:7797–9.
[9] Lal AK, Milton MD. Tetrahedron Lett 2014;55:1810–4.
[10] Lee MH, Giap TV, Kim SH, Lee YH, Kang CL, Kim JS. Chem Commun 2010;46:1407–9.
[11] Azadbakht R, Khanabadi J. Tetrahedron 2013;69:3206–11.
[12] Wang M, Yan FY, Zou Y, Chen L, Yang N, Zhou XG. Sens Actuators B 2014;192:512–21.
[13] Khatua S, Schmittel M. Org Lett 2013;15:4422–5.
[14] Chen HH, Guan RF, Cao DX, Liu ZQ, Sun YH, Ma L, Wang KG, Shan YY. Mater Lett 2014;122:70–3.
[15] Zhang YY, Wu M, Wang YQ, He XW, Li WY, Feng XZ. Talanta 2013;117:196–202.
[16] Kwon WS, Lee G, Do S, Joo T, Rhee SW. Small 2014;10:506–13.
[17] Zhang SR, Wang Q, Tian GH, Ge HG. Mater Lett 2014;115:233–6.
[18] Li Y, Zhao Y, Cheng H, Hu Y, Shi G, Dai L, Qu L. J Am Chem Soc 2012;134:15–8.
[19] Zhang LQ, Dai S, Li L, SL. J Am Chem Soc 2012;134:18932–5.
[20] Sun HJ, Gao N, Wu L, Ren JS, We WL, Qu XG. Chem Eur J 2013;19:13362–8.
[21] Wang FX, Gu ZY, Lei W, Wang WJ, Xia XF, Hao QL. Sens Actuators B 2014;190:516–22.